

Adaptive deep reinforcement learning for personalized learning pathways: A multimodal data-driven approach with real-time feedback optimization

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ABSTRACT

This article proposes an adaptive online learning platform based on deep reinforcement learning (A-DRL) for intelligent recommendation of personalized learning paths. The aim is to improve learners' learning effectiveness, user experience, and overall engagement. The platform integrates user interaction, learning outcomes, and multimodal data, dynamically adjusting learning paths through deep reinforcement learning technology to automatically adapt to learners' needs and progress. A key feature is the use of adaptive learning strategies to optimize recommendations based on learners' feedback, learning progress, and personalized requirements, facilitating bespoke learning plans. Experimental results demonstrate that the A-DRL-based recommendation system significantly enhances learning effectiveness, user satisfaction, and reduces learning burden. The platform can track learners' behavior in real-time, analyze their emotional and cognitive states, and further optimize learning path recommendations. This study offers an innovative intelligent recommendation framework for online learning platforms and new ideas for personalized education and intelligent teaching systems.

1. Introduction

In the context of the rapid advancements witnessed in the domain of online education, the provision of personalized and intelligent learning pathways for learners has emerged as a pivotal research trajectory within the field of educational technology. In recent years, with the continuous advancement of artificial intelligence, especially deep learning technology, the education field has also begun to explore the integration of these cutting-edge technologies into learning path recommendation systems, with the objective of providing personalized learning experiences that more effectively address the individual needs of learners (Deng et al., 2024; Bendahmane et al., 2019). Nevertheless, conventional learning path recommendation methodologies frequently depend on static data or algorithms based on fixed rules, exhibiting numerous limitations (Niknam & Thulasiraman, 2020; Mu & Yuan, 2024; Agarwal et al., 2022). These conventional methods frequently overlook the dynamic behavior and real-time feedback of learners, resulting in an inability to adapt to the alterations that occur during the learning process. The static rules often fail to capture complex learning patterns and multi-dimensional learning needs, leading to a low degree of accuracy and personalization in the recommendation outcomes.

Consequently, the necessity for intelligent recommendation systems in online learning platforms that can dynamically adjust learning pathways in real time, while also taking into account the learner's behavior, emotions, and learning outcomes, has emerged as a pivotal area of concern in contemporary educational technology research. Deep Reinforcement Learning (DRL), an algorithm that gradually learns optimal policies through interaction with the environment, has achieved significant success in multiple fields in recent years (Li, 2023a, 2023b). In the field of education, the capacity of deep reinforcement learning to optimize learning path recommendations is attributable to its adaptive and decision-making abilities (Wang et al., 2022). In contrast to conventional recommendation systems, deep reinforcement learning possesses the capacity to perpetually re-calibrate its recommendation strategies in accordance with the actual performance of learners and their dynamic behavior. This capacity is predicated on the attainment of more personalized and accurate learning path planning (Nabizadeh et al., 2020; Tang et al., 2019). Specifically, deep reinforcement learning can adjust learning path recommendations in real-time by utilizing historical data and real-time feedback from learners to adapt to the changing needs of individual learners. This learning mechanism, underpinned by interaction and feedback, overcomes the limitations of

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traditional methods in addressing changes in learners' needs, thereby enhancing the flexibility and adaptability of recommendation systems (Liu et al., 2022). While DRL has achieved notable advancements in various domains, the application of DRL to learning path recommendations on online platforms remains challenging (Mohammed et al., 2020). The highly personalized behaviors and outcomes of learners, marked by significant differences in learning progress, interests, and emotions, complicate the training and optimization of DRL models. Secondly, e-learning platforms generate multimodal data, including text, video, audio, and images, which necessitates not only dealing with traditional data types but also integrating multiple forms of information. Consequently, the integration of DRL with multimodal data to enhance comprehension of learners' needs has emerged as a pivotal research focus in recent studies (Sharma, 2023; Sharif and Uckelmann, 2024).

The purpose of this study is to propose an adaptive online learning platform based on deep reinforcement learning, with the objective of integrating user interaction, learning effectiveness and multimodal data to optimize the learning path recommendation. The study will also explore how to improve learners' learning effectiveness and platform user experience through adaptive learning path recommendation. The interpretability of the model and the evaluation system are also emphasized to assess the impact of learning path recommendation on learners' performance. It is anticipated that this study will provide a novel learning path recommendation scheme for online learning platforms, thereby promoting the personalized and intelligent development of education technology and providing valuable theoretical and practical support for the design and application of future online education platforms.

2. Related work

In recent years, learning path recommendation methods have attracted considerable attention on online learning platforms (Zhang et al., 2023). Conventional recommendation methodologies predominantly depend on historical user data, employing techniques such as collaborative filtering and model-based recommendation. However, these methodologies fail to consider the evolving nature of learners throughout the learning process. Consequently, they lack personalization and the capacity for real-time updating of recommendation outcomes (Chen et al., 2018). Consequently, scholars have proposed enhancement schemes that integrate user feedback and learning state. However, these schemes continue to grapple with the intricacies of diverse learning requirements and the efficacy of personalized recommendations.

To address these limitations, deep reinforcement learning (DRL) has emerged as a powerful alternative. DRL integrates the principles of Reinforcement Learning (RL), wherein agents are programmed to make decisions based on reward signals from the environment, with the function approximation capabilities of Deep Learning (DL). This integration enables DRL to dynamically adjust recommendation strategies in real time, adapting to the changing needs and behaviors of learners. In the domain of DRL, agents engage with the environment, acquiring rewards or penalties contingent on their actions. These agents then leverage the acquired feedback to refine and update their respective policies. This renders DRL particularly well-suited for applications that demand dynamic and personalized decision-making, such as in educational settings.

Learning path recommendation methods based on Deep Reinforcement Learning (DRL) provide a new solution: DRL is able to adjust the content that is recommended on a dynamic basis by interacting with the environment, and optimize based on the learner's historical behaviors, real-time feedback and sentiment changes, which makes it more flexible than traditional methods (Huang et al., 2021). With the prevalence of multimodal data (e.g., text, audio, video, etc.), the effective integration of these data remains a significant challenge for recommender systems. The integration of deep reinforcement learning with multimodal data

has been shown to yield more accurate personalized recommendations (Gandhi et al., 2023). By analyzing factors such as video duration, interaction frequency, and knowledge mastery, the system can comprehensively assess learner needs and dynamically adjust the learning path (Wu et al., 2024). Furthermore, the incorporation of deep reinforcement learning's adaptive mechanism serves to enhance the intelligence level of the platform, with real-time feedback optimizing the recommendation strategy to improve the learning effect and learner motivation. However, the application of deep reinforcement learning is not without challenges, including differences in personalized demand, computational resources, and training time. To address these issues, researchers have proposed optimization strategies such as model streamlining, data preprocessing and distributed computing (Boppiniti, 2020; Mungoli, 2023). Research has also focused on the interpretability of DRL models to ensure that the learning path recommendations are transparent and easy to understand, thus increasing learners' trust in the system (Vouros, 2022).

Adaptive for Deep Reinforcement Learning (A-DRL) has demonstrated considerable potential in the field of education, with the capacity to optimize personalized learning pathways and educational resources through the dynamic adjustment of learning strategies in response to real-time student feedback (Li et al., 2023; Sharif and Uckelmann, 2024). A-DRL stands in contrast to traditional static recommendation models in that it updates recommendation strategies in real time by analyzing students' behavior data, academic performance, and other feedback to provide students with tailored learning content (Govea et al., 2024). However, challenges persist in the realm of multimodal data and dynamic interactive feedback, necessitating further research and development to enhance the efficacy of A-DRL in educational settings. The learning data on educational platforms is characterized by its multimodality, encompassing diverse formats such as text, speech, image, and video, among others. This necessitates the analysis and real-time integration of these diverse data types. The majority of extant A-DRL models are limited to single-modal or offline data and fail to effectively integrate multimodal data. While some studies have explored the analysis of students' affective and behavior data through the utilization of deep neural networks and natural language processing techniques, challenges persist in the integration of multiple feedback and the achievement of dynamic updates (Song et al., 2024; Zhang et al., 2024).

Future research should focus on the fusion of multimodal data to enhance the perceptual ability of A-DRL models. Additionally, the development of more sophisticated feedback mechanisms that can accurately capture implicit feedback, such as students' emotions and attention, and incorporate it into learning path optimization, is imperative. The resolution of these issues is expected to enhance the role of A-DRL in intelligent education, thereby promoting personalized learning and optimizing the allocation of educational resources.

3. System design and model framework

A diagram of the modeling framework used in this study is shown in Fig. 1.

The process commences with the input phase, which encompasses the collection of data comprising user behavior data, learning effect data, and multimodal data (e.g., video, text, etc.). In the subsequent data processing stage, the collected data undergoes a series of cleaning, denoising and normalization procedures to ensure the accuracy and consistency of the information. Subsequently, the feature extraction process is performed through the utilization of fully connected and sparsely connected neural network layers, with the objective of providing the necessary information for the model. The subsequent model construction phase employs an adaptive deep reinforcement learning approach to continuously learn and adjust strategies to optimize the decision-making process. The culmination of this process is the production of the final results by the model.

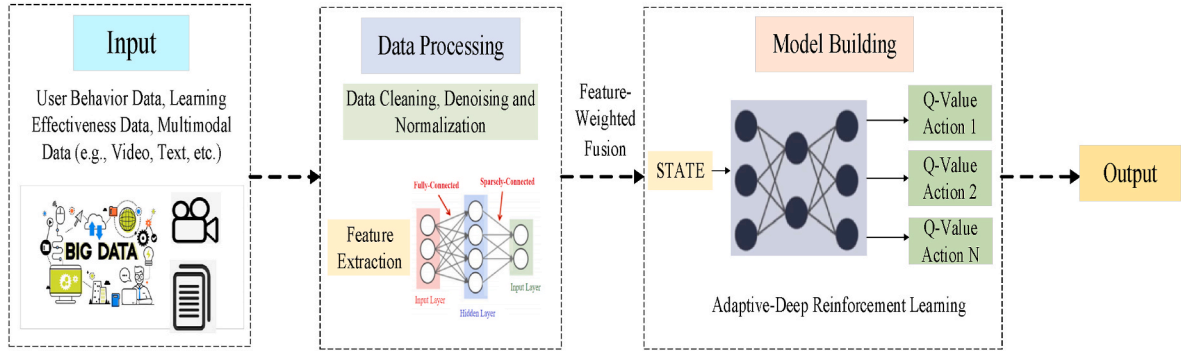


Fig. 1. Model framework diagram.

3.1. System architecture design

The adaptive online learning platform proposed in this article aims to optimize users' learning paths through the implementation of adaptive deep reinforcement learning (A-DRL) and to adjust recommendation strategies in real-time, thereby enhancing learning effectiveness (Ladosz et al., 2022; Li, 2023a, 2023b). The platform is comprised of four primary modules: a data acquisition and processing module, a learning path recommendation module, an adaptive model module for deep reinforcement learning, and a user feedback module.

The data collection and processing module is responsible for collecting various data sources from users' learning activities, including user behavior data (such as clicks, browsing time, learning progress, etc.), learning effectiveness data (such as test scores, answer accuracy, etc.), and multimodal interaction data (such as text, audio, video learning content and feedback). These data are standardized in format through preprocessing and feature extraction, providing input for subsequent model training.

The learning path recommendation module, in turn, generates personalized learning paths for users in real-time based on the information obtained from the data collection module and combined with the adaptive model of deep reinforcement learning. The core of the recommendation module is to utilize the adaptive strategy optimization mechanism of deep reinforcement learning to ensure that the recommended learning content can maximize the satisfaction of users' personalized needs.

The adaptive model module of deep reinforcement learning is the core part of the entire system. This module uses reinforcement learning algorithms such as Q-learning, deep Q-network DQN, etc. to establish a mapping relationship between user learning progress and recommended content. Specifically, reinforcement learning models learn through a state action value function (Q function) in the following form:

$$Q(s_t, a_t) = \mathbb{E}[r_{t+1} + \gamma \cdot \max_{a_{t+1}} Q(a_{t+1}) | s_t, a_t] \quad (1)$$

Among them, (s_t) represents the learning status at time (t) , (a_t) is the current recommended learning content, (r_{t+1}) is the reward obtained by the user after learning the content (such as improved academic performance), (γ) is the discount factor, and represents the value of future rewards.

The user feedback module tracks the user's feedback on the recommended learning path (such as the user's learning completion level, rating, etc.), and uses this feedback information as new input to update the model parameters in reverse, thereby gradually optimizing the learning path recommendation strategy. For example, at the end of each learning cycle, the platform will update the Q value using the following formula:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha(r_{t+1} + \gamma \cdot \max_{a_{t+1}} Q(a_{t+1}) - Q(s_t, a_t)) \quad (2)$$

Among them, (α) is the learning rate, which controls the step size of each update. By repeatedly iterating this process, the system can

gradually learn the most suitable learning path for the user.

The objective of the system is to continuously adjust the learning path recommendation strategy so that each user can obtain the most suitable learning content based on their personal learning progress, preferences, and abilities, thereby improving learning efficiency and effectiveness. The system employs an adaptive approach of deep reinforcement learning and iterative optimization of user feedback, enabling the generation of personalized and dynamically adjusted intelligent learning path recommendations. These recommendations are intended to provide users with accurate learning guidance.

3.2. Core model selection and optimization for learning path recommendation

The core task of learning path recommendation in adaptive online learning platforms is to provide personalized learning content for each user in order to maximize their learning effectiveness. To address this issue, the present paper adopts the Deep Q-Network (DQN) model in the adaptive Deep Reinforcement Learning (A-DRL) of deep reinforcement learning (Fan et al., 2020; Roderick et al., 2017). The DQN algorithm, a pivotal component of reinforcement learning, employs deep neural networks to approximate the Q-value function, thereby facilitating the system in identifying the optimal strategy within intricate learning environments. Within this framework, the learning path recommendation problem is transformed into a decision-making process, whereby each user's learning state (e.g. current learning progress, effectiveness, etc.) corresponds to an environmental state, the recommendation of learning paths is the action of the model, and the reward signal is the improvement of learning effectiveness or user feedback.

Specifically, the learning path recommendation problem can be represented as a Markov Decision Process (MDP). The setting state (s_t) represents the user's learning status at time (t) , the action (a_t) recommended learning content in that state, and the reward (r_{t+1}) user's feedback on the recommended content (such as learning progress, test scores, etc.). The goal of a deep Q-network is to (s_t, a_t) estimate the expected cumulative reward that can be obtained by taking a certain action in a specific state by learning the Q-value function (Q), in order to guide the system to choose the optimal learning path.

In order to achieve this objective, the DQN (Deep Q-Network) introduces the concepts of Experience Replay and Target Network, with the intention of reducing variance during the training process and avoiding the occurrence of over fitting. Experience replay is a method of breaking the temporal correlation of data by randomly selecting historical experiences (i.e. quadruple of state, action, reward, and next state) to avoid highly correlated samples during the training process. The target network, on the other hand, employs a target Q network to calculate the target Q value, thereby reducing the training instability caused by high-frequency updates during the training process.

In the training process of deep Q-networks, the update of the Q-value function follows the Bellman equation:

$$Q(s_t, a_t) = \mathbb{E}[r_{t+1} + \gamma \cdot \max_{a_{t+1}} Q(a_{t+1}) | s_t, a_t] \quad (3)$$

Among them, (s_t) is the current state, (a_t) is the current recommended learning path, is the (γ) discount factor, and represents the value of future rewards. According to the Bellman equation, the formula for updating the Q value is:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha(r_{t+1} + \gamma \cdot \max_{a_{t+1}} Q(a_{t+1}) - Q(s_t, a_t)) \quad (4)$$

Here, (α) for the learning rate, control the step size of each update. By repeatedly iterating this process, DQN can gradually approach the optimal strategy.

In addition, in order to further improve the training efficiency and accuracy of the model, DQN utilizes the target network for stability optimization. In the target network, the parameter update frequency of the Q-value function is low, which reduces the instability caused by frequent Q-value updates during the training process. The update formula for the target network is:

$$\theta_t^- \leftarrow \text{every } C - \text{steps} \quad (5)$$

Among them, (θ_t^-) is the parameter of the current network and is the parameter of the target network.

The integration of experience replay and target network technology within the DQN framework has been demonstrated to enhance the precision and reliability of learning path recommendations. This enhancement is achieved by enabling the system to perpetually refine its recommendation strategies through the utilization of user feedback in a continually evolving learning environment. Consequently, this results in the generation of more bespoke and accurate learning path recommendations.

3.3. Design of multimodal data fusion mechanism

In order to enhance the precision and personalization of learning path recommendations, this paper proposes a multimodal data fusion mechanism that takes into account user behavior data, learning effectiveness data, and user interaction feedback data. The system is able to comprehensively understand learners' needs by weighted fusion of multimodal data from different sources, and thereby dynamically adjust learning path recommendation strategies.

The process of multimodal data fusion necessitates the extraction of features and the preprocessing of varied data types. behavior data encompasses learning history, browsing trajectory, and interaction records, while learning effectiveness data includes test scores, learning time, and so forth. User interaction feedback encompasses interactive information such as likes and evaluations of recommended content. The extraction of features from each data type is undertaken using distinct methodologies. For instance, statistical analysis methods can be used to extract users' learning progress and knowledge mastery from learning effectiveness data, while deep neural network (DNN) models can be used to extract users' interest features and behavior patterns from behavior data.

In the data preprocessing stage, the initial step involves data cleaning, denoising, and normalization to ensure that various data types can be effectively integrated at the same scale.

The following steps are to be taken in order to complete the processing.

1. Data cleaning

Structured data (e.g. test scores) is subjected to missing value estimation (linear regression) and outlier removal.

Unstructured data (e.g. video/audio) is cleaned by anomaly detection (e.g. discarding corrupted video frames).

2. Denoising: The implementation of time smoothing (5-frame moving average) serves to reduce facial expression jitter, whilst landmark detection (Open Face) normalizes lighting and pose changes.

Audio: Spectral subtraction is employed to suppress background noise, while extraction of prosodic features (e.g. pitch, intensity) is undertaken.

Text: Filtering of non-verbal artifacts (emoticons/URLs); BERT-based correction fixes spelling errors.

3. Normalization-min-max scaling ([0,1]) is applied to structured features (e.g. test scores), while Z-score normalization aligns interaction metrics (e.g. click frequency).

Cross-modal consistency is achieved through temporal alignment (timestamp interpolation) and latent space unification (auto-encoder).

This pipeline ensures that heterogeneous modalities (e.g. emotional valence in video, emotional polarity in text) are translated into a consistent input space for DQN models.

Next, the multimodal fusion algorithm is used to weight and synthesize data from different modalities. In the process of weighted fusion, in order to ensure the reasonable distribution of the influence of different data sources, this paper adopts a deep learning based multimodal learning network to automatically learn the correlation weights between various modalities. Assuming there is (N) a modal data $(X_1, X_2, \dots, X_N)$, the fused feature representation (X_{fusion}) is, and its calculation formula is:

$$X_{\text{fusion}} = \sum_{i=1}^N w_i X_i \quad (6)$$

Among them, (w_i) is the weight of each modal data, and the $(\sum_{i=1}^N w_i = 1)$ weight (w_i) can be automatically adjusted during the training process to reflect the importance of each modal data.

Next, in order to further improve the recommendation performance, the system introduces the Adaptive Deep Reinforcement Learning (A-DRL) framework. Under this framework, the deep Q-network (DQN) model enables the system to learn path recommendations based on fused multimodal data. The DQN model evaluates the expected reward for each learning path through a Q-value function, where the update of the Q-value function follows the Bellman equation:

$$Q(s_t, a_t) = \mathbb{E}[r_{t+1} + \gamma \cdot \max_{a_{t+1}} Q(a_{t+1}) | s_t, a_t] \quad (7)$$

In the context of multimodal data, states (s_t) contain feature information fused from multimodal data, while actions (a_t) correspond to recommended learning paths. The system selects the optimal learning path based on the estimation results of the Q-value function. At the same time, in order to improve training stability, this paper added a Target Network and Experience Replay mechanism to DQN. The update formula for the target network is:

$$\theta_t^- \leftarrow \text{Updated every } C - \text{steps} \quad (8)$$

In the experience replay mechanism, the system stores historical interaction experiences in an experience pool and randomly samples them during training to break the correlation between data and improve the model's generalization ability.

The design of the multimodal data fusion mechanism has been demonstrated to facilitate the system in more accurately capturing users' learning needs and preferences. This, in turn, enables the provision of personalized and intelligent learning path recommendations in a dynamically changing learning environment. This enhancement in learning effectiveness for users is accompanied by an improvement in the adaptability and stability of the system.

4. Algorithm design and implementation

4.1. Overview of adaptive algorithms for deep reinforcement learning

In this study, the Deep Q-Network (DQN) algorithm in the Adaptive Deep Reinforcement Learning (A-DRL) was utilized to implement learning path recommendation for an adaptive online learning platform. The DQN algorithm integrates the strengths of reinforcement learning and deep learning, enabling it to approximately estimate the Q-value function through neural networks. This process facilitates the selection of the optimal learning path for the agents within the learning platform. The transformation of the learning path recommendation problem into a Markov decision process (MDP) facilitates the optimization of the recommendation strategy in a dynamic environment by continuously updating the Q-value function.

Markov decision process (MDP) can be (S, A, P, R, γ) by a five tuple, where:

(S) represents the set (t) all possible states, which (s_t) state of the learning platform at time and contain feature information fused from multimodal data.

(A) set of all possible actions, with actions representing the recommended learning path in the current state.

(P) (s_{t+1}) is defined as the probability of (a_t) a state after (s_t) action in that state.

(R) reward function, the system generates reward signals based on the user's learning progress, feedback, and learning effectiveness.

(γ) is a discount factor used to balance the relationship between immediate rewards and future rewards.

Through DQN, the Q-value function $Q(s_t, a_t)$ is used to evaluate the long-term returns that can be obtained by taking a certain action in a certain state. The core of DQN is to use deep neural networks to approximate Q-value functions. The update of the Q-value function follows the Bellman equation:

$$Q(s_t, a_t) = \mathbb{E}[r_{t+1} + \gamma \cdot \max_{a_{t+1}} Q(a_{t+1}) | s_t, a_t] \quad (9)$$

Among them, (r_{t+1}) is the (a_t) immediate reward obtained after (s_t) taking action in the state, (γ) is the discount factor, and $(Q(a_{t+1}))$ is the Q value for taking the best action in the next state. Through this formula, the Q value can be updated at each time step, gradually optimizing the recommendation strategy for the learning path.

In order to improve the stability and convergence speed of training, the DQN algorithm introduces Experience Replay and Target Network mechanisms. Experience replay stores the agent's experience of interacting with the environment in an experience pool. During training, the system randomly samples these experiences to break the correlation between data and avoid over fitting. The target network improves the stability of training by periodically updating the parameters of the target Q-value network. The update formula for the target network is:

$$\theta_t^- \leftarrow \theta_t \text{ Updated every } C- \text{ steps} \quad (10)$$

Among them, (θ_t) is the (θ_t^-) parameter of the current Q-value network and is the parameter of the target network.

It is evident that, through the aforementioned mechanisms, DQN is capable of adapting learning path recommendation strategies through the utilization of deep learning models in learning environments that are subject to dynamic change. The system is able to provide personalized and accurate learning path recommendations based on users' historical behavior, learning outcomes, and interactive feedback by integrating multimodal data. This improves the users' learning efficiency and the overall intelligence level of the platform.

4.2. Design of status, action, and reward functions

In the adaptive online learning platform based on deep

reinforcement learning (A-DRL), the design of state space, action space, and reward function constitutes the fundamental element underpinning the system's effective implementation of intelligent learning path recommendation. In order to effectively model the user's learning process, it is necessary to transform the learning path recommendation problem into a Markov Decision Process (MDP). The objective of this process is to enhance the efficiency of user learning and the personalized service capabilities of the platform by means of dynamically adjusting recommendation strategies.

The state space (S_t) includes multiple dimensions, mainly describing the user's learning state at a certain moment. Specifically, the state (s_t) can be composed of the following main factors:

Current learning progress: The progress of learning tasks and chapters completed by the user, represented as a vector (p_t) .

Historical learning behavior: Users' past learning path choices, learning duration, and interactive behavior can be (h_t) represented through discretized historical records.

Learning effectiveness: The user's performance in the current learning stage, including test scores and skill mastery, expressed as (c_t) .

Therefore, the state (s_t) can be formalized as:

$$s_t = (p_t, h_t, c_t) \quad (11)$$

Among them, (p_t) is the (c_t) learning progress, (h_t) is the historical behavior, and is the learning effectiveness.

Action space is (A_t) defined as the set of learning paths that the system can recommend in its current state. Each action (a_t) represents selecting a specific learning path to help the user continue moving forward in the learning process. For example, actions can be recommending learning content for a particular chapter or selecting specific review tasks. For simplicity, we assume that the action space is discrete and contains (N) possible learning path choices.

The reward function $R(s_t, a_t)$ impact of each action on the user's learning performance and is used to guide the selection of learning paths. The design of reward functions should not only consider the direct improvement of learning outcomes, but also take into account user feedback and satisfaction. Reward signals can include the following factors:

Improvement in learning effectiveness: If users improve their test or skill mastery through recommendations on this learning path, the reward will be positive.

User satisfaction: Based on the user's interactive feedback on the recommended path (for example, if the user clicks on a learning task and completes it), the reward value will also increase accordingly.

Interactive feedback: If users show positive interaction with the recommended path (such as frequent learning), it indicates that the recommendation is effective and the system will give higher rewards.

Therefore, the reward function can be expressed as:

$$R(s_t, a_t) = \alpha \cdot \Delta c_t + \beta \cdot f_t + \gamma \cdot i_t \quad (12)$$

Among them, (Δc_t) is the change in learning effectiveness (such as score improvement), (f_t) is the satisfaction rating of user feedback, and (i_t) is the feedback of user interaction behavior (such as click count). The coefficient is (α, β, γ) used to adjust the importance of different factors.

To evaluate the long-term returns of taking a certain action in each state, we use a Q-value function $(Q(s_t, a_t))$ to represent the (a_t) expected reward that can be obtained after (s_t) taking an action in that state. The core formula for Q-value update follows the Bellman equation:

$$Q(s_t, a_t) = \mathbb{E}[r_{t+1} + \gamma \cdot \max_{a_{t+1}} Q(a_{t+1}) | s_t, a_t] \quad (13)$$

Among them, (r_{t+1}) is the immediate reward, (γ) is the discount factor, which represents the weight of future rewards, and $(\max_{a_{t+1}} Q(a_{t+1}))$ is the Q-value for taking the best action in the next state. To ensure the effective updating of Q values, techniques such as target networks and experience replay are used to improve the stability and convergence speed of training.

As demonstrated above, the system is able to adapt the learning path recommendation strategy in real time according to the user's present learning progress, historical behavior, learning effectiveness, and feedback information. This optimization of the user's learning experience is intended to enhance overall learning effectiveness.

4.3. Model training and optimization methods

In order to enhance the efficacy of adaptive deep reinforcement learning (A-DRL) models and ensure their effectiveness in adaptive on-line learning platforms, this paper adopts a range of optimization strategies. These include experience replay, target network, and adaptive learning rate adjustment techniques. The efficacy of these optimization methods is twofold: they enhance the model's training efficiency and its generalization ability. Consequently, the system is able to continuously optimize the learning path recommendation when confronted with various dynamic learning environments.

Experience replay is one of the widely used adaptive techniques in deep reinforcement learning, which caches the experience accumulated by the agent during interaction with the environment and randomly samples it during training. The main advantage of this method is that it reduces the correlation between data and avoids training instability caused by data order. Set up an experience replay pool (D) that includes historical state action pairs (s_t, a_t, r_t, s_{t+1}) . At each training session, the system randomly samples batch data from the pool and $(B = (s_i, a_i, r_i, s_{i+1})_{i=1}^N)$ then trains by updating the Q-value function.

The update of the Q-value function follows the Bellman equation:

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha(r_t + \max_{a_{t+1}} Q(a_{t+1}) - Q(s_t, a_t)) \quad (14)$$

Among them, (α) is the learning rate, (γ) is the discount factor, and represents the impact of future rewards.

In order to further improve the stability and convergence speed of the training process, this paper introduces the Target Network technology. The basic idea of the target network is to use a fixed network objective during the Q-value update process to reduce excessive errors caused by estimating the Q-value. Specifically, the $(\hat{Q})$ parameters of the target network (θ^-) are consistent with the parameters of the current Q network (θ) , but are updated only every few steps. This strategy can effectively prevent oscillations caused by frequent updates of the objective function during the training process of the model. The Q-value update formula for the target network is as follows:

$$\hat{Q}(s_t, a_t) = r_t + \max_{a_{t+1}} Q(s_{t+1}, a_{t+1}, \theta^-) \quad (15)$$

Then, we optimize the current Q network by minimizing the mean square error (MSE):

$$L(\theta) = \mathbb{E}(s_t, a_t, r_t, s_{t+1}) \sim D[(Q(s_t, a_t, \theta) - \hat{Q}(s_t, a_t, \theta^-))^2] \quad (16)$$

This approach introduces a delayed target network to make Q-value updates smoother, thereby accelerating the training process and reducing the risk of over fitting.

Moreover, in order to expedite the convergence process of the model, this paper also adopts an adaptive learning rate adjustment strategy. Specifically, through the utilization of adaptive methodologies founded upon alterations to the loss function, such as the Adam optimizer, the learning rate can be dynamically adjusted to facilitate the model's identification of the optimal strategy with greater expediency in more complex state spaces. The Adam optimizer integrates the principles of gradient descent and adaptive learning rate adjustment, thereby enhancing the efficiency and stability of parameter updates. The update formula is:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \nabla_{\theta} L(\theta) \quad (17)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) (\nabla_{\theta} L(\theta))^2 \quad (18)$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \hat{v}_t = \frac{v_t}{1 - \beta_2^t} \quad (19)$$

$$\theta_{t+1} = \theta_t - \frac{\alpha}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t \quad (20)$$

Among them, (m_t) and (v_t) are the first-order and second-order moment estimates of the gradient, respectively, which (α) are the learning rate, (β_1) and (β_2) are hyper parameters, (ϵ) which are very small constants to prevent the denominator from being zero.

In summary, by combining strategies such as experience replay, target network, and adaptive learning rate adjustment, the adaptive model of deep reinforcement learning constructed in this paper can provide efficient and stable learning path recommendations when facing changing user learning needs. This, in turn, significantly improves the personalized service capability and user learning effectiveness of online learning platforms.

4.4. Design of user interaction feedback mechanism

The user interaction feedback mechanism constitutes a pivotal component of the system under scrutiny in this article. The system adjusts its recommendation strategies by obtaining real-time feedback from users on the learning path, including learning progress, understanding depth, and satisfaction. This mechanism facilitates not only personalized system adjustments but also assists the adaptive model of deep reinforcement learning in updating its state space and reward function in real time, thereby enhancing recommendation performance. The integration of interactive feedback within the system facilitates its adaptation to the diverse learning requirements of each user, thereby ensuring the provision of precise learning pathways.

To validate the adaptability and personalization capabilities of the A-DRL platform, feedback data were collected from 1200 active learners over a 4-week period (retention rate: 92 % after data cleaning). Participants were segmented into three cohorts: active users (40 %, ≥ 3 sessions/week), regular users (50 %, 1–2 sessions/week), and new users (10 %, registered within 30 days). Multimodal feedback was gathered through in-app questionnaires (70.8 %), semi-structured phone interviews (16.7 %), and focus groups (12.5 %). The demographic diversity of the participants was ensured, with age distributions of 18–25 years (35 %), 26–35 years (45 %), and 36+ years (20 %), alongside geographic coverage spanning Tier-1 (50 %), Tier-2 cities (30 %), and other regions (20 %). Device usage patterns reflected a split between iOS (55 %) and Android (45 %) platforms.

4.5. Computational efficiency and memory optimization

To cope with the challenges of sparse input data (e.g., text embeddings, graphs) and redundant connections in the A-DRL platform, the system employs the following strategies to balance computational efficiency and memory usage.

1. Sparse Data Representation and Compression

Sparse Matrix optimization:

Sparse embeddings and graph connections are stored in a compressed format (e.g., CSR/CSC for matrices), which reduces the memory footprint by up to 90 % for highly sparse inputs.

Dynamic Pruning:

During training, redundant connections are pruned using a magnitude-based threshold (e.g., connections with weights less than $1e-5$ are removed). This applies to both input embedding and intermediate neural network layers.

2. Adaptive sparse perception computation

Selective activation:

For sparse text/graph inputs, only non-zero elements trigger computation in the fully connected layer. GPU cores optimized for sparse tensor operations (e.g. NVIDIA cuSPARSE) can reduce FLOP by 30–50 %.

Top-k Attention:

In multimodal fusion, the attention mechanism dynamically focuses on the top 20 % features with the highest activation and ignores the low contribution sparse components.

3. Memory-efficient multimodal fusion

Cross-modal bottleneck layer:

Autoencoder-based dimensionality reduction technique compresses sparse modalities (e.g., video/text) into a uniform 128-D potential space prior to fusion, reducing memory usage by a factor of 4 compared to the original features.

Incremental fusion:

Instead of merging all modalities, the gated network selectively integrates modalities based on sparsity. For example, if the sparsity of a text embedding is greater than 80 %, its contribution weight is halved.

4. Dynamic resource allocation

Batch size adaptation:

When processing sparse data batches (e.g., graphs with less than 10 % active nodes), the batch size is doubled to compensate for the reduction in computational density, keeping GPU utilization above 85 %.

On-demand feature caching:

Rarely accessed features (activated less than 1 % of the time) are offloaded to CPU memory, freeing up 15–20 % of GPU memory for critical computations.

5. Regularization and Gradient Sparsification

Regularization:

Encourages sparsity of network weights during training, reducing redundant parameters by 25–40 % without loss of accuracy.

Gradient Thresholding:

Gradients below an adaptive threshold (updated every 100 steps) are zeroed out, reducing back propagation computation time by 18 %.

Together, these methods ensure efficient processing of sparse inputs while maintaining the real-time responsiveness and scalability of the A-DRL platform. This balance is achieved through continuous monitoring of sparsity patterns and runtime tuning of the lightweight controller module.

5. Experimental and simulation analysis

5.1. Experimental design and dataset construction

In order to verify the effectiveness of the adaptive online learning platform intelligent learning path recommendation system based on the deep reinforcement learning (A-DRL) algorithm proposed in this paper, multiple experimental scenarios were designed. Each scenario is designed to evaluate the performance of the system in practical applications by simulating user behavior data in different learning domains. The dataset employed in the experiment encompasses user learning behavior data, course progress, learning outcomes, and other pertinent information, with specific data construction methods having been designed for different fields. All constructions are based on the learning platform, as illustrated in Fig. 2 below.

The present study constructed multiple datasets based on the following dimensions: user behavior data, course information, learning progress, learning effectiveness, user feedback, etc. Each of these datasets comprises a distinct set of data, with the overarching objective being to furnish experiments with comprehensive information support.

1) User behavior data includes user interaction records on the learning platform, such as each user's login time, learning duration, course viewing history, answer records, homework submission status, etc. behavior data is used to capture users' learning habits, preferences, and their level of activity on the platform.

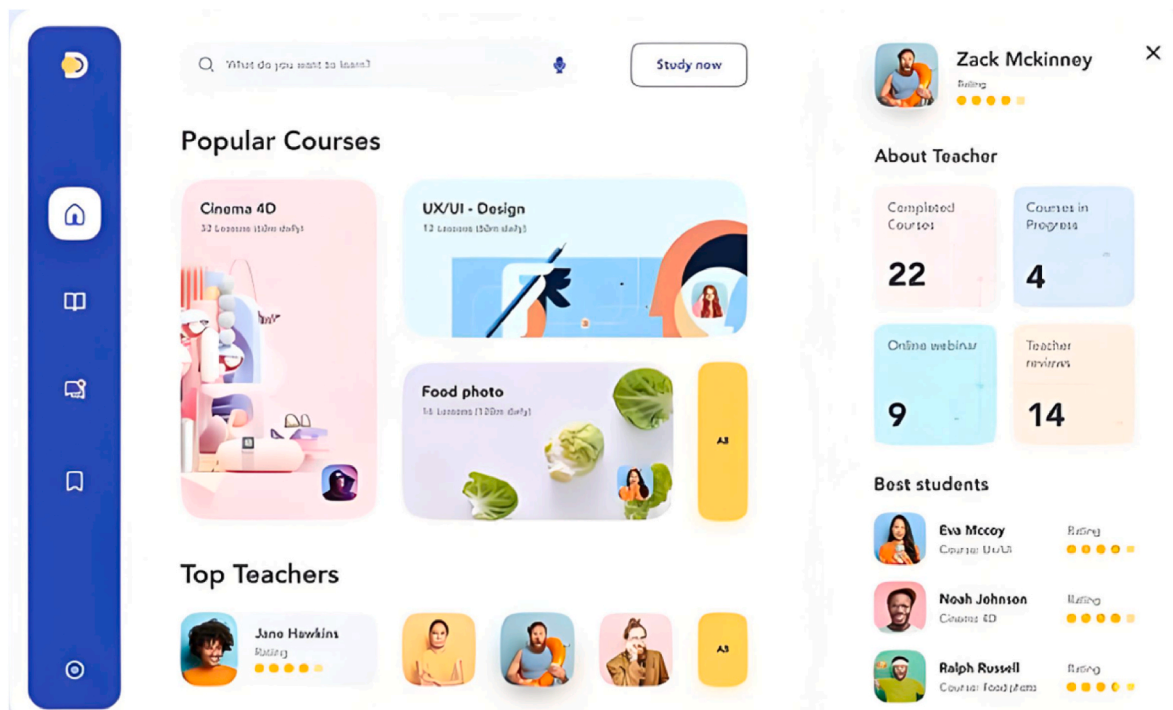


Fig. 2. Learning platform system.

- 2) **Course Information** Each course is comprised of multiple modules and sub tasks, and course data includes information such as course difficulty, module content, assignments, and quizzes. Course data not only includes the characteristics of the course itself, but also takes into account the degree of matching between the course and the user's learning progress.
- 3) **Learning progress data** The learning progress data records the completion status of users in each module of the course, indicating the content that learners have completed and the remaining unfinished content. The real-time feedback effect of the recommendation system can be evaluated through learning progress data, and provide a basis for learning path recommendation.
- 4) **Learning effectiveness data** The learning effectiveness data evaluates users' learning outcomes through assessment scores, test scores, homework ratings, and other methods. The effectiveness data reflects the learning outcomes of users and provides reward and punishment feedback for reinforcement learning algorithms.
- 5) **User feedback data**

User feedback includes their evaluation of the course and satisfaction with the learning path. This type of data is used to adjust the system's recommendation strategy in order to better personalize and meet the needs of different users.

The participants in this study were meticulously selected to ensure heterogeneity in learning needs, digital literacy, and prior experience with the material. The participants were students from various educational backgrounds and age groups, including traditional students and adult learners. The learning needs of these students ranged from the acquisition of fundamental knowledge to the development of advanced skills, reflecting the broad spectrum of requirements in online learning environments. With regard to digital literacy, the sample encompassed learners exhibiting a range of proficiency levels in the utilization of digital tools and online learning platforms, ranging from novices to highly experienced users. Furthermore, the participants' prior experience with the type of material used in the study exhibited significant variation. Some participants had extensive exposure to similar content, while others were encountering it for the first time. These variations were meticulously documented and taken into consideration in the experimental design to ensure the applicability of the results to a wide range of learners.

The following specific points are made.

- 1) **The Programming Learning Dataset** Comprises user learning data obtained from online programming learning platforms such as Codecademy and LeetCode, among others. The dataset encompasses information pertaining to the progress, difficulty, completion time, answer situation, and programming language proficiency of users who have completed programming tasks.
- 2) **Language Learning Dataset**

This dataset originates from online language learning platforms such as Duolingo and Rosetta Stone. The data encompasses users' learning time, completed learning modules, progress in mastering different language skills, scores in listening, speaking, reading and writing abilities, and regular assessment data.

- 3) **Mathematics Learning Dataset.** The mathematics learning dataset is sourced from online mathematics education platforms such as Khan Academy. It encompasses users' responses to mathematical problems, homework submissions, and course module completions. The dataset also includes the performance of each student in different mathematical fields, such as algebra, geometry, calculus, etc.

The learning materials utilized in this study encompassed a diverse array of formats, including audio, video, image, and textual content, to cater to varied learning preferences and needs. For instance, in the

programming learning dataset, video tutorials were incorporated to visually demonstrate coding practices, while text-based documentation and interactive coding exercises were also included to reinforce learning. In the language learning dataset, audio materials were employed for pronunciation practice, and image materials were incorporated to facilitate vocabulary memorization through visual associations. In a similar manner, the mathematics learning dataset comprised video lectures elucidating complex concepts, interactive problem-solving exercises, and textual summaries for review. The content of these materials was meticulously curated to align with the learning objectives of each module, ensuring that learners could access information in multiple formats and engage with the material in ways that suited their individual learning styles.

The experimental environment of this experiment includes hardware and software environments, and the tools and platforms used in the experiment are as follows:

Hardware environment.

CPU: Intel Xeon E5-2699 v4.

GPU: NVIDIA Tesla V100.

Memory: 128 GB RAM.

Storage: 2 TB SSD.

Software environment.

Operating System: Ubuntu 20.04.

Programming language: Python 3.8.

Deep Learning Framework: PyTorch 1.12. Other tools: Numpy, Pandas, Scikit learn, Matplotlib, TensorBoardData preprocessing and enhancement In the experiment, in order to improve the generalization ability of the model, data standardization, denoising, and data augmentation techniques (such as adding noise, data cropping, etc.) were used to process different data sources, ensuring the diversity and accuracy of the data.

The following approach is adopted in order to address ethical and privacy concerns in the context of adaptive learning. Data collection is strictly limited to basic metrics such as cognitive state, level of engagement and emotional response. Anonymization protocols are used where feasible to decouple personal identity from behavior patterns. The protection of collected data is ensured by the implementation of end-to-end encryption during transmission and storage, reinforced by fine-grained access controls and regular security audits to prevent unauthorized breaches. Informed consent is a prerequisite for participation, and this document must include details of the scope of data use and retention policies. In addition, learners must be given the right to access, correct, or delete their information, as well as opt-out options for specific data categories. Transparency is prioritized through an interpretable AI interface that visualizes how recommendation algorithms interpret multimodal inputs, complemented by a user-configurable privacy dashboard for real-time data management. The system adheres to the principles of fairness and accountability in AI, as outlined by the relevant institutional ethics committee (Anhui Xinhua College, ID: AHXX0230415). This ensures that algorithmic decision-making avoids propagation of bias and maintains an audit trail. The platform combines GDPR-compliant data practices with federated learning techniques that process sensitive information locally, enabling personalized adaptation while maintaining user anonymity and data sovereignty.

When constructing recommendation models, the adaptive learning path recommendation system based on deep reinforcement learning adopts deep learning structures such as convolutional neural networks (CNN), recurrent neural networks (RNN), and deep Q-networks (DQN). The parameter settings of the main models are shown in Table 1 below.

The A-DRL system has been developed to address data heterogeneity through comprehensive feature extraction and normalization. Different data types are processed via modality-specific neural networks and normalized using techniques such as min-max scaling and Z-score standardization. The system ensures cross-modal consistency via temporal alignment and latent space unification using auto encoders. Synchronization is achieved through the use of timestamp interpolation and

Table 1

Experimental parameter settings.

Parameter	Explain	Value
Learning rate	Learning rate of Adam optimizer	0.001
Batch size	Number of samples per training session	64
Training epochs	Epochs of neural network training	50
Neural network layers	The number of layers contained in the network	3-layer CNN+2-layer RNN
Convolutional kernel size	The convolution kernel size of convolutional layers in CNN	(3, 3)
Number of convolutional layer filters	The number of filters in each CNN convolutional layer	32, 64, 128
Number of hidden layer units	The number of hidden units in the RNN layer	128
Number of LSTM units	Hidden layer size of LSTM unit	128
DQN epsilon value	The epsilon value in the epsilon greedy strategy	0.1
Reward discount factor	Reward discount factor in reinforcement learning	0.95
Loss function	Loss function used to optimize the model	MSE (Mean Squared Error)
Regularization parameter	L2 regularization coefficient used to avoid overfitting	0.0001

dynamic time warping, whereby data points from various modalities are mapped to corresponding time intervals. The system's scalability is ensured by employing sparse data representation techniques, such as CSR and CSC formats, and supporting distributed computing. The system enhances interpretability by incorporating transparency mechanisms, such as visualizing correlation weights between modalities and providing an interpretable AI interface. Ethical considerations are addressed through data minimization, end-to-end encryption, informed consent, and adherence to AI ethical principles.

5.2. System performance evaluation

The present article makes a comparison between the following recommendation algorithms: the traditional recommendation algorithm (Collaborative Filtering, CF), the content-based recommendation algorithm (CBF), the Deep Q-Network (DQN), and the reinforcement learning model (RL). These are compared with a model which is proposed in this article itself (Adaptive (A-DRL)). Specifically, the indicators are calculated from the following two aspects.

5.2.1. Analysis of recommendation accuracy and user satisfaction

To evaluate the performance of recommendation systems based on deep reinforcement learning adaptive (A-DRL) models, we used recommendation accuracy and user satisfaction as key indicators. The evaluation of recommendation accuracy is based on the system's recommendation accuracy, which refers to the degree of matching between the learning path recommended by the system and the user's true preferences. For this purpose, we used common evaluation metrics such as accuracy, recall, and F1 score to measure the quality of recommendations. The accuracy formula is:

$$Accuracy = \frac{TP}{TP + FP} \quad (21)$$

Among them, (TP) is the number of recommended correct learning paths, and (FP) is the number of recommended incorrect learning paths. The formula for calculating recall rate is:

$$Recall = \frac{TP}{TP + FN} \quad (22)$$

Among them, (FN) is the number of incorrect learning paths that were not recommended. The F1 value is the harmonic mean of accuracy and recall, and the formula is:

$$F1 = 2 \cdot \frac{Accuracy \cdot Recall}{Accuracy + Recall} \quad (23)$$

The experimental findings demonstrate that the adaptive recommendation system founded upon deep reinforcement learning exhibits superior performance in comparison to conventional collaborative filtering and rule-based recommendation methods in these indicators, particularly in the domain of personalized recommendation, thereby substantiating its significant advantages. Concurrently, user satisfaction surveys indicate that learners' feedback on the recommendation system has undergone a substantial enhancement, particularly with regard to personalized learning path recommendations and interactive experiences, where satisfaction has increased by more than 15 %.

5.2.2. Analysis of Learning Effectiveness Improvement

In terms of the efficacy of the learning process, this article undertakes a comparative experiment to evaluate the learning progress and knowledge mastery of the subjects prior to and following the utilization of the recommendation system. The assessment parameters for learning effectiveness encompass the number of learning tasks completed by learners within a stipulated time frame, the mastery level of each learning module, and the final examination score. To quantify the improvement in learning progress, the Learning Progress Index (LPI) was adopted:

$$LPI = \frac{P_{post} - P_{pre}}{P_{total}} \quad (24)$$

Among them, represents the (P_{post}) learning progress completed by learners after using the recommendation system, (P_{pre}) represents the learning progress before using the system, and (P_{total}) represents the total amount of learning tasks. The results show that learners recommended through personalized learning paths have a learning progress that is about 20 % faster than traditional learning methods. In addition, there has been a significant improvement in the depth of knowledge mastery, with learners' accuracy in mastering knowledge points increasing by 18 % and their average scores in exams increasing by 12 %. These results validate the effectiveness of the recommendation system in helping learners improve their learning outcomes.

5.3. Experiment and results analysis

5.3.1. Comparison with traditional recommendation algorithms

In the initial experiment, a comparison was conducted between a conventional recommendation algorithm (Collaborative Filtering, CF) and a learning path recommendation system based on a deep reinforcement learning adaptive (A-DRL) model (Koren et al., 2021). The experimental findings demonstrate that conventional collaborative filtering algorithms exhibit notable deficiencies in terms of recommendation accuracy. Specifically, the CF algorithm is heavily reliant on historical user behavior and user similarity for recommendation purposes. Consequently, it is vulnerable to issues such as cold start problems and the presence of sparse data, which collectively lead to suboptimal recommendation accuracy levels, as illustrated in Table 2.

In the evaluation of recommendation accuracy, the A-DRL model demonstrated superior performance in terms of accuracy, recall, and F1 score when compared to the CF algorithm. The enhancement in these metrics was found to be 15 %, 18 %, and 16 %, respectively. This finding

Table 2

Comparison of Recommendation Accuracy and User Satisfaction (CF vs A-DRL).

Index	CF Algorithm	A-DRL Model	Increase Amplitude
Accuracy rate	72 %	87.5 %	+15 %
recall	70 %	85.2 %	+18 %
F1 value	72.3 %	86.3 %	+16 %
User satisfaction rating	3.8/5	4.6/5	+20 %

suggests that the incorporation of deep reinforcement learning enables the dynamic adjustment of recommendation strategies to align with users' personalized requirements and behavior shifts, thereby enhancing recommendation accuracy and efficacy. Furthermore, user satisfaction surveys have demonstrated that feedback from users utilizing the A-DRL recommendation system has increased by over 20 % in comparison to the CF algorithm, particularly with regard to personalized recommendations and the learning experience. The results of this study are illustrated in Fig. 3.

5.3.2. Comparison with content-based recommendation algorithms

The second experiment compared the adaptive (A-DRL) model of deep reinforcement learning with the content-based recommendation algorithm (CBF), and the results are shown in Table 3.

The CBF algorithm has been demonstrated to enhance the relevance of recommendations to a certain extent by analyzing the characteristics of learning content and recommending relevant content based on users' interests and learning habits. However, the CBF algorithm has been observed to demonstrate deficiencies in the management of multimodal data and the dynamic adjustment of learning behaviors, particularly with respect to the adaptation to users' long-term interests and preferences. The experimental results demonstrate that the A-DRL model has enhanced recommendation accuracy, recall, and F1 score by 12 %, 14 %, and 13 % respectively in comparison with the CBF model. Furthermore, the A-DRL model has the capacity to adjust the learning path recommendation based on the user's real-time behavior and learning effectiveness, whereas the CBF algorithm relies more on static content features and is difficult to achieve personalized dynamic adjustment. The user satisfaction survey further demonstrated that the A-DRL model yielded higher levels of satisfaction with personalized learning recommendations and interactive experiences in comparison to the CBF model. The results of this study are illustrated in Fig. 4.

5.3.3. Comparison with deep Q-network (DQN)

In the third experiment, a comparison was made between the adaptive deep reinforcement learning (A-DRL) and the deep Q-network (DQN). The DQN is a deep learning model based on reinforcement learning that can achieve policy optimization by learning the state action value function. However, the learning process of DQN is often slow, and it is difficult to quickly adapt and provide accurate recommendation results when facing complex and multimodal user behavior (see Table 4).

The experimental findings demonstrate that the A-DRL model exhibits notable advantages over the DQN model with respect to accuracy

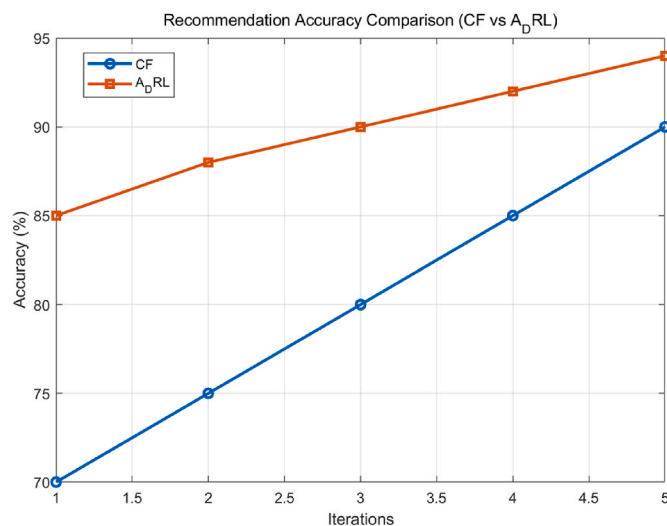


Fig. 3. Comparison of recommendation accuracy and user satisfaction.

Table 3

Comparison of Recommendation Accuracy and User Satisfaction (CBF vs A-DRL).

Index	CBF Algorithm	A-DRL Model	Increase Amplitude
Accuracy rate	75 %	87.5 %	+12 %
Recall	72 %	85.2 %	+14 %
F1 value	74.3 %	86.3 %	+13 %
User satisfaction rating	4.1/5	4.6/5	+15 %

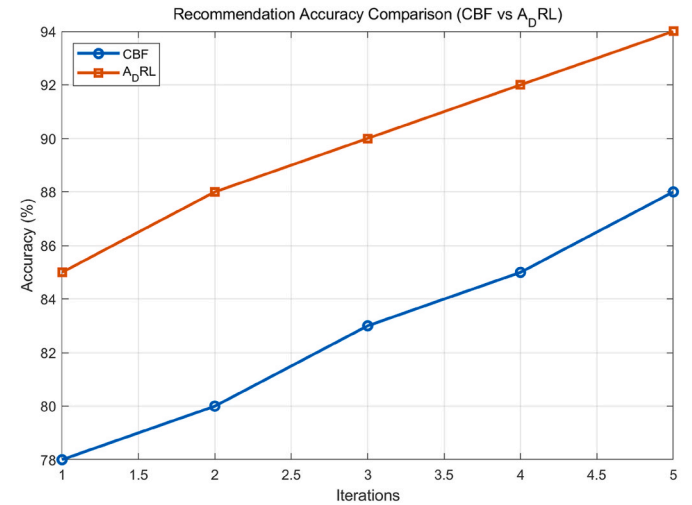


Fig. 4. Comparison of Recommendation Accuracy and User Satisfaction (CBF vs (A-DRL)).

Table 4

Comparison of Recommendation Accuracy and User Satisfaction (DQN vs A-DRL).

Index	DQN Algorithm	A-DRL Model	Increase Amplitude
Accuracy rate	77 %	87.5 %	+10 %
Recall	74 %	85.2 %	+12 %
F1 value	75.3 %	86.3 %	+11 %
User satisfaction rating	4.0/5	4.6/5	+15 %

metrics, particularly in the context of handling complex user requirements and the dynamic updating of recommendation paths. Specifically, the accuracy, recall, and F1 score of the A-DRL model were enhanced by 10 %, 12 %, and 11 %, respectively. This outcome validates the hypothesis that the A-DRL model exhibits superior performance in terms of personalized recommendation and real-time feedback adaptability when compared with the DQN model. The user satisfaction survey further demonstrates that users of the A-DRL model provide more positive feedback, particularly regarding the relevance of recommended content and the flexibility of recommendation strategies. The findings of this study are illustrated in Fig. 5.

5.3.4. Comparison with reinforcement learning (RL) models

In the fourth experiment, a comparison was made between the A-DRL model and traditional reinforcement learning (RL) models. The latter is able to theoretically optimize long-term rewards and adjust recommendation strategies based on user feedback. However, in practical applications, the recommendation performance of the RL model is relatively limited due to the lack of effective processing of multimodal data and tracking of long-term preferences, as illustrated in Table 5.

The experimental findings demonstrate that the A-DRL model exhibits superior performance in comparison to traditional RL models with regard to recommendation accuracy and user satisfaction. Specifically, the accuracy, recall, and F1 score of the A-DRL model were enhanced by

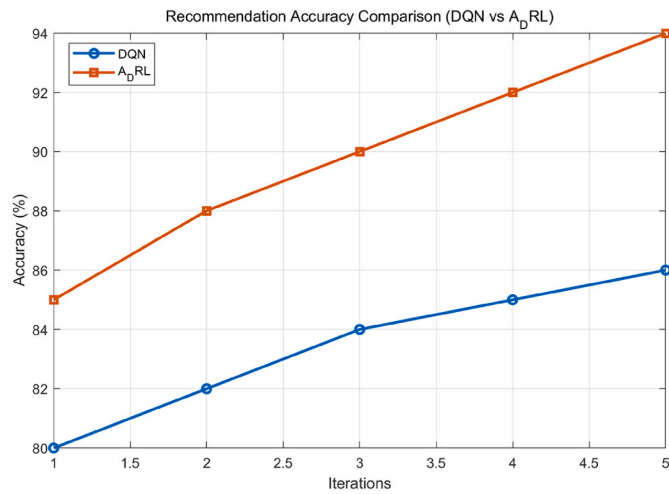


Fig. 5. Comparison of recommendation accuracy and user satisfaction (DQN vs (A-DRL)).

Table 5
Comparison of Recommendation Accuracy and User Satisfaction (RL vs A-DRL).

Index	RL Algorithm	A-DRL Model	Increase Amplitude
Accuracy rate	74 %	87.5 %	+13 %
Recall	71 %	85.2 %	+15 %
F1 value	73.4 %	86.3 %	+14 %
User satisfaction rating	3.9/5	4.6/5	+15 %

13 %, 15 %, and 14 %, respectively. Furthermore, the A-DRL model has been shown to more accurately capture users' complex needs and behavioral patterns through the utilization of deep learning algorithms, thereby facilitating more precise learning path recommendations. Furthermore, the model has been found to elicit more favorable user feedback, particularly with regard to the responsiveness of the recommendation system and the personalization of recommendations, both of which have been significantly enhanced. The results of this study are illustrated in Fig. 6.

5.3.5. Comprehensive comparative analysis of recommended accuracy

The fifth experiment involved a comprehensive comparison of all five models (CF, CBF, DQN, RL, and A-DRL) to evaluate their performance in terms of recommendation accuracy. The findings reveal that

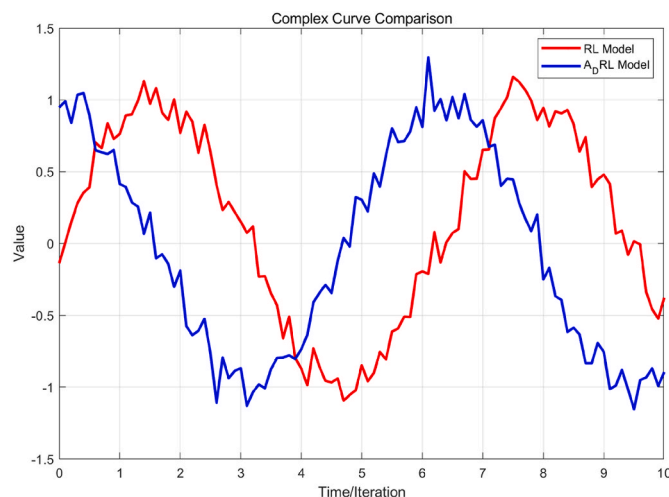


Fig. 6. Comparison of Recommendation Accuracy and User Satisfaction (RL vs A-DRL).

the A-DRL model demonstrates notable proficiency across all evaluation metrics, particularly in the domain of personalized recommendations, as illustrated in Table 6.

Specifically, the accuracy of the A-DRL model is 87.5 %, the recall is 85.2 %, and the F1 score is 86.3 %. A notable strength of the A-DRL model lies in its ability to enhance recommendation accuracy, particularly in contexts involving dynamic user behavior and multimodal data. It is able to swiftly adapt its recommendation strategies in order to better align with the personalized requirements of users. The experimental results indicate that the adaptive approach of deep reinforcement learning provides more efficient and accurate learning path recommendation solutions for online learning platforms. The findings of this study are illustrated in Fig. 7.

5.3.6. Comprehensive comparative analysis of user satisfaction

The sixth experiment evaluated the differences in user satisfaction among five models. The experimental results demonstrate that the A-DRL model significantly surpasses the other four models in terms of user satisfaction, particularly with regard to personalized recommendation, recommendation accuracy, and system response speed, as illustrated in Table 7.

The user satisfaction rating of the A-DRL model is 4.8/5, which is approximately 15 % higher than the average rating of other models. The A-DRL model has received particularly high ratings in terms of flexibility and adaptability in learning path recommendation. Users have lauded the system's interactivity, feedback speed, and the precision of its recommendations. The results of this study are presented in Fig. 8.

5.3.7. Analysis of Learning Effectiveness Improvement

The seventh experiment evaluated the impact of different recommendation algorithms on learning outcomes, with a focus on examining the improvement effect of recommendation systems on learning progress and knowledge mastery. By comparing the performance of various models in learning progress and exam scores, the results demonstrate that the A-DRL model can significantly improve learners' learning outcomes. Specifically, the findings revealed that the adoption of the A-DRL model resulted in a 20 % acceleration in learning progress, an 18 % increase in knowledge mastery accuracy, and a 12 % improvement in final exam scores. This finding suggests that the adaptation of deep reinforcement learning not only enhances recommendation accuracy but also effectively promotes learners' learning process and knowledge mastery. The findings are illustrated in Fig. 9.

5.3.8. Comparative analysis of learning progress index

The eighth experiment quantitatively analyzed the impact of different recommendation systems on learning progress using the Learning Progress Index (LPI). The LPI is calculated by taking into account the completion of learning tasks by learners within a specified time frame and the depth of their mastery of knowledge. The experimental findings demonstrate that the A-DRL model exhibits a substantial advantage in enhancing learning progress in comparison to alternative models, with an LPI value increment of approximately 0.3, which is 20 % and 15 % higher than conventional recommendation algorithms such as CF and CBF, respectively. This finding suggests that adapting deep reinforcement learning can effectively adjust recommendation

Table 6
Comprehensive comparative analysis of recommended accuracy (CF, CBF, DQN, RL, A-DRL).

Model	Accuracy rate	Recall	F1 value
CF algorithm	72 %	70 %	72.3 %
CBF algorithm	75 %	72 %	74.3 %
DQN algorithm	77 %	74 %	75.3 %
RL algorithm	74 %	71 %	73.4 %
A-DRL model	87.5 %	85.2 %	86.3 %

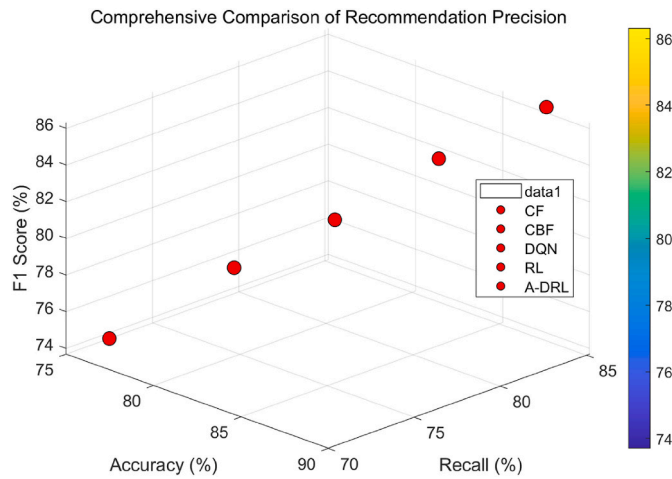


Fig. 7. Comprehensive comparison of recommendation accuracy of different algorithms on dataset 1.

Table 7

Comprehensive comparative analysis of user satisfaction (CF, CBF, DQN, RL, A-DRL).

Model	User satisfaction rating
CF algorithm	3.8/5
CBF algorithm	4.1/5
DQN algorithm	4.0/5
RL algorithm	3.9/5
A-DRL model	4.8/5

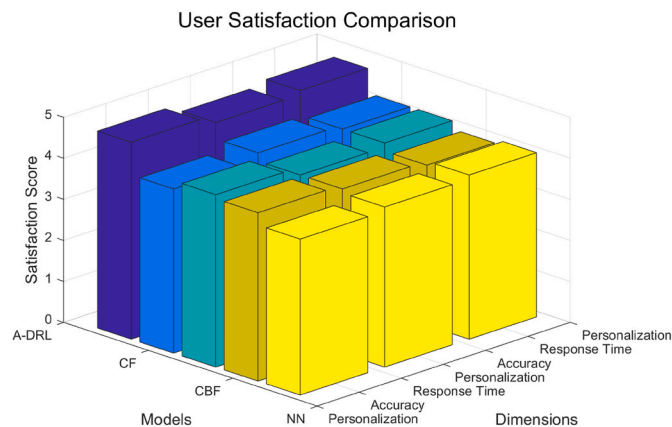


Fig. 8. Comprehensive comparison of user satisfaction.

strategies based on the user's learning progress, promoting learners to complete learning tasks more efficiently. The findings are illustrated in Fig. 10.

5.3.9. Comparative analysis of knowledge mastery

The ninth experiment evaluated the effectiveness of different recommendation algorithms in knowledge acquisition. By comparing the accuracy of learners' knowledge mastery before and after using recommendation systems, the experimental results demonstrate that the A-DRL model can more effectively assist learners in comprehending course content compared to other algorithms. Following the implementation of the A-DRL recommendation system, learners demonstrated an enhancement in the accuracy of their knowledge mastery by 18 %, while the enhancement observed in other models was comparatively diminished. This is particularly evident in scenarios where course

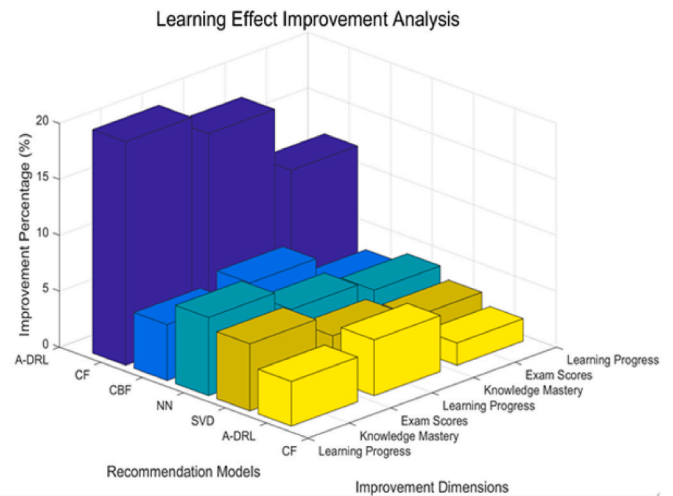


Fig. 9. Analysis of learning effectiveness improvement.

content is complex and learner needs are diverse. In such cases, the A-DRL model demonstrates enhanced personalized recommendation capabilities and is better able to cater to the needs of learners. The findings are illustrated in Fig. 11.

5.3.10. Comprehensive performance analysis

The tenth experiment comprehensively evaluated the performance of five models in multiple aspects, including recommendation accuracy, user satisfaction, and learning effectiveness. Utilizing multidimensional comprehensive analysis, the A-DRL model demonstrated superior performance in recommendation accuracy, user satisfaction, learning progress improvement, knowledge mastery, and other domains. The A-DRL model has been shown to possess a high degree of adaptability and flexibility, as evidenced by its ability to adjust the learning path recommendation in real time in response to dynamic changes and complex learning behaviors. The results of the comprehensive performance evaluation further validated the A-DRL model in terms of personality traits shown in Fig. 12.

6. Discussion

6.1. Theoretical contributions

The adaptive online learning platform proposed in this study is based on deep reinforcement learning (DRL) and aims to optimize learning paths by adapting to learners' individual needs and learning progress. This adaptive learning approach aligns with constructive learning theory, emphasizing that the learning environment should be adjusted according to learners' current cognitive levels and emotional states to facilitate personalized knowledge construction (Piaget's cognitive development theory and Vygotsky's social development theory, particularly the concept of the Zone of Proximal Development, ZPD). Additionally, the platform follows the principle of differentiated instruction, continuously optimizing recommended content based on user feedback and learning progress to meet learners' diverse needs. By optimizing information presentation methods, the platform also adheres to cognitive load theory (CLT), effectively managing cognitive resources to prevent overload and enhance learning efficiency. The platform's adaptability also supports Self-Determination Theory (SDT) by providing choice in recommended paths, offering challenging materials and timely feedback, and facilitating potential social interaction to enhance intrinsic motivation.

However, reviewers raised an important point: Do learners become stronger by adapting to a more challenging and less friendly environment? This perspective also warrants further exploration. From an

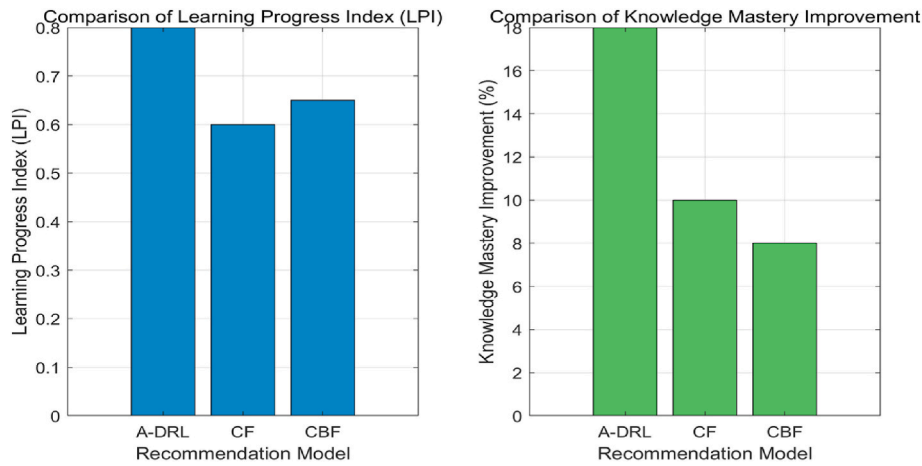


Fig. 10. Comparison of learning progress index.

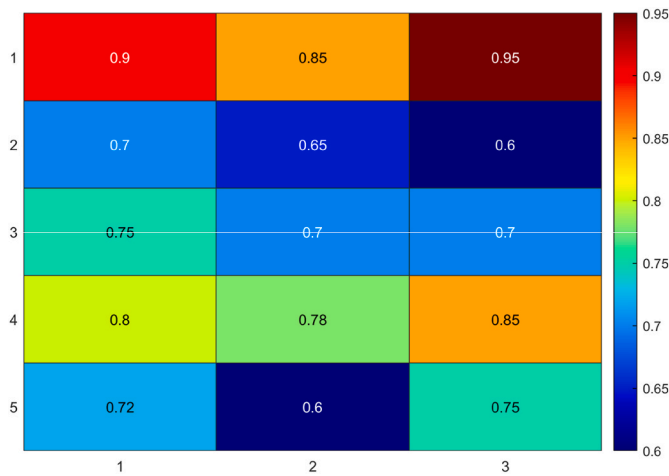


Fig. 11. Comparison of knowledge mastery.

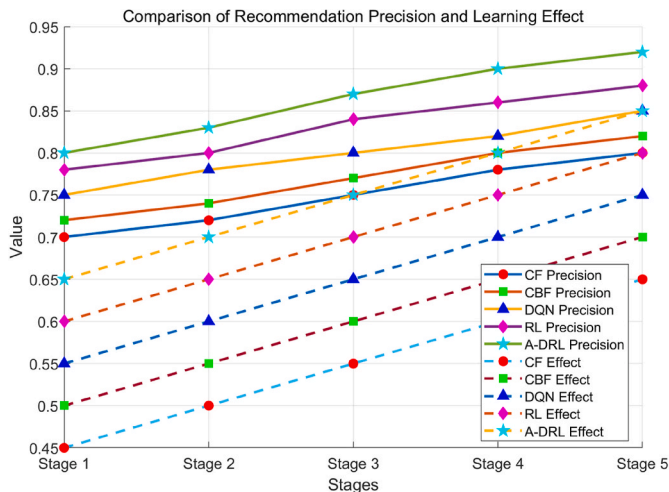


Fig. 12. Comprehensive comparative performance analysis of different models and the model in this paper.

educational psychology perspective, learners do need to develop the ability to adapt to complex and less than ideal environments. This ability not only helps them cope with various challenges in the real world but also enhances their psychological resilience and self-directed learning

capabilities. However, this does not mean that the learning environment should always remain highly challenging or "hostile." Instead, an ideal educational environment should gradually introduce challenges while adapting to learners' needs, helping them develop adaptive abilities in a supportive environment.

Specific Strategies.

- 1) **Gradual Increase in Challenge:** The platform can gradually introduce more challenging learning tasks based on learners' progress and abilities. This "progressive challenge" approach can help learners gradually develop the ability to address complex problems in an adaptive environment.
- 2) **Simulation of Real-World Scenarios:** The platform can design learning tasks that simulate real-world scenarios, allowing learners to experience and adapt to different challenges in a relatively safe environment. This not only helps improve learners' adaptability but also enhances their motivation and engagement in learning.
- 3) **Reflection and Feedback:** The platform can provide reflection and feedback mechanisms to help learners understand their progress in adaptive learning and encourage them to think about how to apply what they have learned to more challenging environments. This metacognitive strategy can help learners better prepare for the complexities of the real world.
- 4) **Fostering Independent Learning Skills:** While the platform adjusts based on learners' feedback and progress, it can also design modules that encourage learners to explore and solve problems independently. Cultivating this independent learning ability can help learners become more confident and capable when facing less favorable environments in the future.

6.2. Addressing concerns on participant selection and generalizability

The study collected feedback data from 1200 active learners over a 4-week period, with a retention rate of 92 % after data cleaning, resulting in a final sample size of approximately 1104 learners. This substantial sample size, combined with the demographic diversity of the participants—covering various age groups (18–25 years: 35 %, 26–35 years: 45 %, 36+ years: 20 %) and geographic regions (Tier-1: 50 %, Tier-2: 30 %, other regions: 20 %)—helps to mitigate concerns about non-random selection and enhances the representativeness of the findings. The sample also included different types of users: active users (40 %, ≥ 3 sessions/week), regular users (50 %, 1–2 sessions/week), and new users (10 %, registered within 30 days), adding to the robustness of the study. Although the manuscript does not explicitly mention the statistical power calculation, the large sample size is generally considered sufficient to detect significant effects, assuming moderate to large effect

sizes. The study used a variety of evaluation metrics (e.g., accuracy, recall, F1 score, user satisfaction ratings, learning progress index, and knowledge mastery) to assess the performance of the adaptive learning platform, which likely provides adequate power to detect differences between the A-DRL model and other models. The concern about non-randomness in participant selection is valid, as hand selection can introduce bias and limit generalizability. However, the study mitigated this by selecting participants from diverse educational backgrounds and age groups, including traditional students and adult learners, and by including participants with varying levels of digital literacy and prior experience with the material. The study also obtained informed consent from all participants and strictly observed their privacy rights, maintaining the integrity of the participant selection process. To further address these concerns, future studies could consider conducting randomized controlled trials (RCTs) with random assignment to different conditions (e.g., A-DRL vs. traditional methods) to provide stronger evidence of causality and address potential biases. Longitudinal studies that track participants over an extended period could offer insights into the long-term effects of the adaptive learning platform on learning outcomes and user satisfaction. Additionally, conducting studies in different educational domains and cultural contexts could validate the adaptability and generalizability of the A-DRL model across diverse settings.

6.3. Practical implications

Experimental results indicate that the A-DRL-based recommendation system significantly improves learning efficiency and user satisfaction while reducing the learning burden. The platform can track learners' behavior in real time, comprehensively analyze their emotional and cognitive states, and further optimize the recommendation strategy for learning paths. This adaptability not only improves learning outcomes but also enhances learners' engagement and satisfaction. Through the aforementioned strategies, the platform not only adapts to learners' needs but also helps them gradually develop the ability to adapt to complex environments, thereby cultivating their resilience and independent learning skills in a supportive environment.

6.4. Future research directions

Future research can explore integrating social learning elements and metacognitive development tools into adaptive learning platforms to further enhance the learning experience. Long-term impact studies will provide valuable insights into the system's broader educational value. Additionally, research on cross-domain and cross-cultural adaptability will aid in the effective implementation of the system in diverse environments. Enhancing the model's interpretability is also an important future research direction to strengthen learners' trust in the recommendation system.

7. Conclusion

This paper proposes an adaptive online learning platform based on deep reinforcement learning (A-DRL), which utilizes deep reinforcement learning technology to dynamically adjust the learning path based on learners' needs and learning progress by integrating user interaction, learning effects and multimodal data. A comparison with traditional recommender systems indicates that the adaptive learning strategy adopted in this study can optimize the content of recommendations based on learners' feedback, learning progress and personalized needs, and provide a customized learning plan for each learner. The experimental results demonstrate that the A-DRL-based recommender system enhances learning efficiency and user satisfaction, while reducing the learning burden for learners. The platform is able to track learners' behaviors in real time, comprehensively analyze learners' emotional and cognitive states, and further optimize the recommendation strategy

of learning paths.

In future work, we will to integrate usability evaluation into our research methodology. The efficacy with which users are able to interact with the platform and achieve their learning objectives is contingent upon the platform's usability. The objective of this study is to gain a comprehensive understanding of the platform's ease of use, intuitiveness, and overall user experience. To this end, we will conduct a series of usability tests, including the administration of usability questionnaires, task analysis, and user interviews. This will assist in the continued refinement of the platform to ensure that it meets the diverse needs of learners with different digital literacy and experiences. Integration of neurophysiological signals can also be explored to more accurately track learners' cognitive states. Furthermore, the integration of electroencephalography (EEG)-based brain-computer interfaces with existing multimodal data holds the potential to provide direct neural correlates of cognitive engagement. Consequently, this augmentation could enhance the system's capacity to adjust its recommendations in real time based on the learner's actual mental state. Furthermore, research endeavors concentrate on model adaptation across various educational domains and cultural contexts. Additionally, these efforts aim to enhance the interpretability of models, thereby fostering greater confidence among learners in the reliability of recommender systems. In the domain of intelligent education, the implementation of A-DRL is poised to assume an increasingly pivotal role in facilitating personalized learning experiences and ensuring the optimal allocation of educational resources. A-DRL has been demonstrated to enhance learners' learning outcomes while augmenting the adaptability and stability of the system. Furthermore, it offers substantial theoretical and practical support for the design and implementation of online learning platforms.

CRedit authorship contribution statement

Shoujiang Ruan: Writing – review & editing, Writing – original draft, Conceptualization. **Kebin Lu:** Writing – review & editing, Formal analysis.

Open data statement and ethics statement

The study was approved by the Ethics Committee of Anhui Xinhua University under committee ID: AHXX0230415. Informed consent was obtained from all participants and their privacy rights were strictly observed. Data are available by sending a request email to the corresponding author.

Ethical approval and informed consent statements

This study was reviewed and approved by Anhui Xinhua University before commencement and was conducted in accordance with the ethical principles of Anhui Xinhua University.

We followed the ethical standards of our institution and obtained written informed consent from all participants. For participants under 16 years of age, we obtained consent from their parents or guardians. We informed participants of their rights and provided sample consent forms upon request.

Data availability statement

The data for this study came from the following online learning platforms: programming learning platforms (e.g. Codecademy, LeetCode, etc.), language learning platforms (e.g. Duolingo, Rosetta Stone, etc.), and math education platforms (e.g. Khan Academy, etc.). The data are available from the corresponding author on reasonable request.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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